Chemometrics and Intelligent Laboratory Systems 200 (2020) 103998

Contents lists available at ScienceDirect

Chemometrics and Intelligent Laboratory Systems

journal homepage: www.elsevier.com/locate/chemometrics

Combination of least absolute shrinkage and selection operator with Bayesian Regularization artificial neural network (LASSO-BR-ANN) for QSAR studies using functional group and molecular docking mixed descriptors

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ARTICLE INFO

Keywords:
Azine derivatives
QSAR
Molecular docking
LASSO
Artificial neural network

ABSTRACT

A combination of least absolute shrinkage and selection operator (LASSO) with Bayesian Regularization feed-forward artificial neural network (LASSO-BR-ANN) was used as a new approach in the quantitative structure-activity relationship (QSAR) studies. A mixture of the docking derived descriptors with the simple functional group (structural) features was also introduced as a new ensemble of descriptors for accurate QSAR modeling. The performance of introduced approaches was tested with QSAR modeling of the biological activities (pEC_{50}) of 73 azine derivatives as new non-nucleoside reverse transcriptase inhibitors (NNRTIs) for treatment of HIV disease. Molecular docking descriptors (MDDs) were generated from ligand-receptor interactions and functional group features derived using Dragon 5.5 software. The dataset was divided into three sets of training, validation, and test data. LASSO, as a penalized regression method, was applied to the training data set for the selection of the most relevant descriptors among the mixture of the structural and MDDs. LASSO selected descriptors were used as inputs in the construction of the Bayesian Regularization artificial neural network (BR-ANN) model. The results showed that the addition of functional group properties to the MDDs improves the accuracy of the model. Under the optimum conditions, LASSO-BR-ANN was successfully applied for the prediction of PEC_{50} values for compounds in the external test set with mean square error (MSE) and coefficient of determination (R^2) values of 0.07 and 0.88, respectively. Some of the prediction statistical parameters of the model were calculated and all of them were in their acceptable ranges, which confirm the validity of the proposed QSAR model.

1. Introduction

Acquired Immunodeficiency Syndrome (AIDS) is an infectious disease, first identified in 1983. AIDS is caused by the human immunodeficiency virus (HIV) that affects the immune cells in the blood (lymphocytes) and in the tissues. Drugs associated with this disease are divided into Nucleoside reverse transcriptase inhibitors (NRTIs), Non-nucleoside reverse transcriptase inhibitors (NNRTIs), Protease inhibitors (PIs), Integrase inhibitors (INSTIs), Fusion inhibitors (FIs), Chemokine receptor antagonists (CCR5 antagonists), Entry inhibitors CD4-directed post-attachment inhibitors [1,2]. Among these compounds, NNRTIs are a key component of highly active antiretroviral therapy (HAART) for long-term management of infection. These inhibitors have unique anti-viral potency, high activity and low toxicity [1] and so far 5 drugs of this type of inhibitors have been approved by the Food and Drug Administration (FDA) as active antiretroviral agents. Recently, azine derivatives (compounds in which the carbon atom (s) of the benzene ring replaced with the nitrogen atom) as new NNRTIs were synthesized and classified as a group of reference drugs of Rilpivirine and Etravirine [3–7]. Azine compounds include derivatives of piperidinylamino-diarylpyrimidine, pyridine (synthesized by scaffold refining of DAPY), and derivatives such as pyridazine and pyridone, which are synthesized by the replacement of the central pyrimidine with pyridazine and pyridone, respectively. All of these derivatives belong to the family of Rilpivirine or Etravirine. In addition to the foregoing, triazine derivatives are another class of azine compounds that are structurally similar to Etravirine reference drugs. Azine derivatives as new NNRTIs have desirable anti-HIV activities compared to the reference

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https://doi.org/10.1016/j.chemolab.2020.103998
Received 25 October 2019; Received in revised form 26 February 2020; Accepted 16 March 2020
Available online 19 March 2020
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drugs [7,8]. Therefore, the design and synthesis of new azine compounds with greater activity or less toxicity are of interest. On the other hand, the study of the relationship between structure and biological activity of drug-like compounds is a powerful tool for improving the targeted design of drugs [9]. Thus, we decided to study the relationship between the structure of various azine derivatives and their anti-HIV activities as a targeted drug design tool for designing new azine derivatives with high biological activities.

Quantitative structure-activity relationship (QSAR) study is one of the computational chemistry methods that play a central role in improving directed drug design by identification of some of the structural characteristics of the molecule, which has a special relationship with biological activity. In addition, QSAR reduces the cost of studies by lowering the empirical studies required for drug design. In QSAR studies, a mathematical model is developed to establish a relation between a set of molecular characteristics (descriptors) extracted from the structure of compounds and biological activities. Based on which structural features are used to develop the QSAR model for drug design purposes, QSAR studies can be categorized into ligand-based drug design (LBDD) and structure (receptor)-based drug design (SBDD) that in many cases, these two methods complement each other. The ligand-based method typically uses a set of compounds with a specific activity. Often, this method is used in cases where there is no structural information about the target protein. The key concept in LBDD methods is that compounds with structural components similar to known active compounds are likely to have activities similar to those of the active compounds. SBDD methods are an essential part of drug design procedure, and knowledge about the biological structure, especially the binding site of the protein, is required in this method. The benefit of the SBDD approach is obtaining the ligand-receptor binding interaction information that is very helpful in chemical design and revision of design to promote activity and/or reduce toxicity [10,11]. The main requirement of structure-based QSAR studies is computational software that can assess the molecular descriptors based on ligand-receptor interactions [12]. However, literature surveys show that the most reported QSAR studies are by means of the ligand-based method [13,14], which has quite limited application domain and is often just used to optimize compounds with the same scaffold [15]. Recently, researchers have made efforts to use ligand-receptor interaction information as descriptors in the QSAR studies [16]. However, in many cases, such descriptors are not included in the final QSAR model as vital descriptors [17–19] and/or the constructed models do not have desirable prediction ability. This can be attributed to the fact that the information obtained from the interaction of the receptor-ligand alone cannot be descriptive enough [20]. Therefore, it seems necessary to use some descriptors containing the information about ligand structure along with those containing ligand-receptor interactions in the QASR studies [10]. There are a large number of structural descriptors that are easily calculated by the commercial software. Among these descriptors, functional group descriptors are interpretable and are calculated by counting the functional groups in the structure of the compounds. In addition, the interaction of the drug-like compounds with target molecules is strongly influenced by the type and number of functional groups in their structures. So, as a new research idea, we decided to combine simple and informative functional group descriptors with those calculated from the ligand-receptor interaction information and use their assembly to build a QSAR model with improved prediction ability.

It has been established in QSAR studies that when the number of descriptors that participate in the final QSAR model is less, the model will be simpler and more comprehensible [13]. In addition, many descriptors do not have a significant relationship with biological activities, and in some cases, there is a pairwise correlation or co-linearity between descriptors, which causes some drastic problems in model construction [21,22]. Therefore, the introduction of new methods for variable selection or investigation of the efficiency of the existing methods in the selection of the most important descriptors is still of interest in QSAR studies. Different classical methods have been used to select the most effective descriptors. These methods include forward and backward selection [23] and stepwise regression [24]. Among these methods, stepwise, which is a regression method based on changes in the residual sum of squares, is more widely used [14,25,26]. The stepwise regression method as a common descriptor selection method in QSAR studies has elementary problems. The high bias in the regression coefficients causes some insignificant variables to be selected while some important variables are discarded, for example, at the probability level of 0.05 ($P_{value} = 0.05$), the chances of choosing a non-variable are about 0.185. Also, in the presence of co-linearity, the efficiency of the method is greatly reduced [27–29]. Due to problems associated with the stepwise method, this approach is not considered by statisticians as an efficient variable selection method. Currently, penalized regression methods have been considered as very efficient variable selection methods [30–33]. Some of these methods include smoothly clipped absolute deviation (SCAD) [34] least absolute shrinkage and selection operator (LASSO) [27] and Elastic Net [35]. In LASSO a penalty term is added to the residual sum of squares (RSS), which controls the model complexity and increases the performance of LASSO in the variable selection procedure with shrinkage of the regression coefficients to zero. This method is less encountered with the over-fitting problem due to the use of the $\ell 1$-norm penalty [30]. Additionally, LASSO as a variable selection method has other advantages such as sparsity and interpretability [27]. According to literature studies, LASSO has frequently been used as a regression method for model building in QSAR studies [31,33,36,37]. There is only one report regarding the use of the LASSO method in the variable selection and reduction of data dimension prior to QSAR model building via support vector machine (SVM) approach [30]. So far there is no report on the performance of LASSO method for variable selection and preparation of data for subsequent QSAR model construction using efficient methods such as artificial neural network (ANN) method. Therefore, further studies are needed to demonstrate the usefulness of LASSO in reducing the dimension of data in QSAR studies and this study can be considered as the first step in evaluating the efficiency of LASSO method in the reduction of data dimension and selection of the most important descriptors for constructing the QSAR model using ANN method.

In this study, a successful QSAR model was developed for the prediction anti-HIV activities in a wide range of azine compounds including DAPYs, Triazine, Pyridazine, and Pyridine. A mixture of descriptors extracted from ligand-receptor interactions with the functional group descriptors was used as a new class of descriptors in the construction of the QSAR model. Combining these two categories of descriptors, in addition to entering the ligand-receptor interaction information in the final model, also incorporates the structural details of the compounds in the QSAR modeling. In addition, this study is the first report on the application of the LASSO method in the selection of the best relevant descriptors prior to QSAR modeling via the ANN method.

2. Theory of LASSO

Consider the following multiple regression model $y = X\beta + \varepsilon$ where $y = (y_1.y_2.....y_n)^T$ is the n-dimensional response vector, $X = (x_1.x_2.....x_n)^T$ is the design matrix of order $n \times p$, in which $x_i = (x_{i1}.x_{i2}.....x_{ip})^T$ is the i-th p-dimensional explanatory vector, $\beta = (\beta_1.\beta_2.....\beta_p)^T$ is the p-dimensional feature vector, known as regression parameters, and $\varepsilon = (\varepsilon_1.\varepsilon_2.....\varepsilon_n)^T$ is the error term. One method of estimating the feature vector is to use the ordinary least squares (OLS) approach in which we minimize the $F(\beta)$ with respect to β, where

$$F(\beta) = \varepsilon^T\varepsilon = (y - X\beta)^T(y - X\beta) \tag{1}$$

The resulting LS estimator of β is then given by

$$\widehat{\beta} = (X^TX)^{-1}X^Ty \tag{2}$$

When we have too many features (large p), we may face some

difficulties in obtaining a good estimator of β using the OLS method. Too many explanatory variables may cause a multicollinearity problem. In this case the variance of $\widehat{\beta}$ is very large and we cannot rely on the estimates. In case $p > n$, the matrix X^TX is near singular and the inverse does not exist. Hence, as a remedy, one may decrease the number of features. In other words, we need to select important variables by using a variable selection method [27].

One fitting approach in which the choice of variables is carried out is the stepwise regression. A variable is considered for addition to or subtraction from the set of explanatory variables in each step based on statistical values like R^2, t-stats, etc. Some of the most commonly used stepwise regression methods are standard stepwise, forward selection, backward elimination. Briefly, the aim of this modeling technique is to maximize the prediction power/accuracy with the minimum number of predictor variables.

Ridge regression improved prediction error by shrinking large regression coefficients in order to reduce over-fitting. On the other hand, it takes care of the multicollinearity problem well. But it does not perform variable selection and so does not help to make the model more interpretable. There is another method that is able to achieve both goals by forcing the sum of the absolute value of the regression coefficients to be less than a fixed value, which forced certain coefficients to be set to zero, effectively choosing a simpler model that does not include those coefficients. It was introduced by Robert Tibshirani in 1996. LASSO penalizes the absolute size of the regression coefficients. In addition, it is capable of reducing the variability and improving the accuracy of linear regression models. It can be defined as:

$$\widehat{\beta} = argmin_{\beta \in \mathbb{R}^p} \left\{ \|y - X\beta\|_2^2 + \lambda \|\beta\|_1 \right\} \tag{3}$$

LASSO regression leads to the penalize (or equivalently constraining the sum of the absolute values of the estimates) values which causes some of the parameter estimates to turn out exactly zero. Larger the penalty applied, further the estimates get shrunk towards absolute zero. This results in variable selection out of given n variables. It is important to say the assumption of LASSO regression is the same as LS regression except normality is not be assumed here. LASSO shrinks the regression coefficients to exactly zero which helps us in feature selection. If the group of predictors are highly correlated, LASSO picks only one of them and shrinks the others to zero [27].

3. Computational methods

3.1. Dataset

A set of data includes 73 compounds of azine derivatives was collected from the literature [3–7]. The main skeletons of the compounds are shown in Fig. 1. The details of the substitutions, the activity values (EC_{50} in μM) and its p-values (pEC_{50} in M), which are used as dependent variables, are given in Table 1. The data set is randomly divided into three sets named training (51 compounds), validation (11 compounds) and test (11 compounds) sets. The selection was conducted in such a way that the validation and test series are an appropriate representation of the training set. In all studies including variable selection and model building, the validation and test data series were eliminated from the total data set. The training set was used for the construction of models and validation and test series were used as the internal and external evaluation of the constructed models, respectively. The information for all the compounds in the data set along with the endpoint values were given in SMILES format in Table S1.

3.2. Drawing and optimization of molecular structures

The molecular structure of each compound was first drowned in the HyperChem 8.0.5 [38] and after adding hydrogen atoms, the three-dimensional structure of the compounds was optimized using the semi-empirical AM1 method. The optimization continued until the root mean square of the energy gradient reached 0.001 kcal mol^{-1}. All calculations were performed at Restricted Hartree-Fock (RHF) calculation level and the optimized structures of the compounds were finally saved in PDB format using ViewerLite software.

3.3. Preparation of the protein and cognate ligand structures

With respect to the relative similarity between ligands present in the active site of proteins (cognate ligand) with the compounds studied and according to the available reports [3–7], crystallographic structures with PDB codes of 3M8Q (for compounds 1 to 43) and 3MEC (for compounds 44 to 73) were selected. The crystallographic structures of these receptors which containing the cognate ligands were downloaded from the Protein Data Bank (PDB) [8]. In the next step, the downloaded files were opened in ViewerLite5.0 software, and water molecules, co-factors, and the corresponding cognate ligands were removed. The modified structures were saved as the structures of receptors in PDB file format. To extract the cognate ligands, the protein files were opened in the ViewerLite5.0 software and all the amino acids (enzyme structure), water molecules and co-factors were removed. The residue structure was stored as the cognate ligand structure in the PDB file format. These structures (receptors and cognates ligands) were manipulated in Autodock 4.2 software [39] for obtaining the optimum ligand-receptor (LR) complex.

3.4. Descriptor generation

MDDs were extracted from the best pose of the ligand in the active site of the receptor (the best LR conformation). The LR conformation with the best pose was defined as a conformer with the more negative binding energy, which is usually observed in the first cluster. The docking parameters were set to the optimal condition derived from the validation procedure (dock-re-dock of the cognate ligand into the receptor active site). 43 molecules (compounds of 1–43) were separately docked into the active site of 3M8Q protein and the remainder 40 molecules docked into the active site of 3MEC protein. The resulted docking outputs were saved in docking log (dlg) file format. Finally, by analyzing the information in each file, the best conformer of LR complexes was identified and 9 MDDs were extracted. The name of descriptors and their abbreviations and meanings are summarized in Table 2. In addition to MDDs, several simple and interpretable structural descriptors were also calculated using Dragon 5.5 software that they incorporate the structural features of the compounds into the QSAR model. The list of structural descriptors that have been involved in the final QSAR model and their symbols are shown in Table 2.

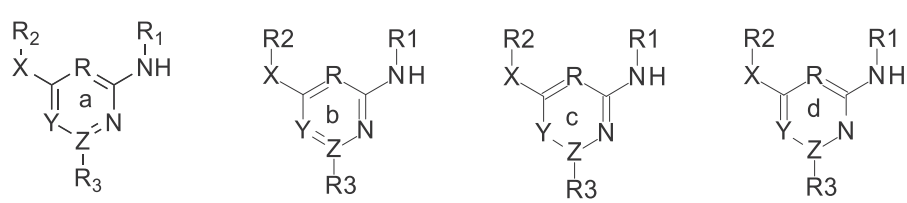

Fig. 1. Main skeletons of the compounds studied.

Table 1
Structural features and biological activities of azine derivatives. X, Y, Z, R, R_1, R_2, and R_3 are defined in Fig. 1.

No.	Y	X	R_1	R_2	R_3	EC_5	pE_{50}	interval
a1^{T}	N	O	piperidin	2,4,6-TriMe-Ph	-NH-4-CN-Ph	0.14±0.05	6.85	6.19–7.52
a2^{IT}	N	O	piperidin	2,6-DiMe-4-CN-Ph	-NH-4-CN-Ph	0.038± 0.0028	7.42	7.28–7.56
a3^{ET}	N	O	p-1-yl- 4-SO_2NH_2-Ph	2,4,6-TriMe-Ph	-NH-4-CN-Ph	0.19±0.04	6.72	6.33–7.11
a4^{T}	N	O	p-1-yl -4-SO_2CH_3-Ph	2,4,6-TriMe-Ph	-NH-4-CN-Ph	0.13±0.08	6.89	5.74–8.04
a5^{T}	N	O	p-1-yl - 4-Pyridyl	2,4,6-TriMe-Ph	-NH-4-CN-Ph	0.24±0.12	6.62	5.69–7.55
a6^{T}	N	O	p-1-yl - 4-SO_2NH_2-Ph	2,6-DiMe-4-CN-Ph	-NH-4-CN-Ph	0.13±0.05	6.89	6.17–7.60
a7^{IT}	N	O	p-1-yl - 4-SO_2CH_3-Ph	2,6-DiMe-4-CN-Ph	-NH-4-CN-Ph	0.058±0.028	7.24	6.34–8.14
a8^{ET}	N	O	p-1-yl -4-Pyridyl	2,6-DiMe-4-CN-Ph	-NH-4-CN-Ph	0.047±0.011	7.33	6.89–7.76
a9^{T}	N	NH	p-1-yl - 4-SO_2 NH_2-Ph	2,4,6-TriMe-Ph	-NH-4-CN-Ph	0.077±0.065	7.11	5.54–8.69
a10^{T}	N	NH	p-1-yl - Pyridin-4-yl	2,4,6-TriMe-Ph	NH_2	0.0049±0.0024	8.31	7.40–9.22
a11 ^{IT}	N	NH	p-1-yl - 4-COOEt-Ph	2,4,6-TriMe-Ph	NH_2	0.0115±0.51	7.94	7.85–8.02
a12^{T}	N	NH	p-1-yl - 4-$CONH_2$-Ph	2,4,6-TriMe-Ph	NH_2	0.02043±0.0197	7.69	5.89–9.49
a13^{T}	N	NH	p-1-yl - 4-SO_2NH_2-Ph	2,4,6-TriMe-Ph	NH_2	0.00575±0.00264	8.24	7.38–9.10
a14^{T}	N	NH	p-1-yl- 4-SO_2Me-Ph	2,4,6-TriMe-Ph	NH_2	0.00555±0.00148	8.26	7.76–8.75
a15^{T}	N	NH	p-1-yl- Phenyl	2,4,6-TriMe-Ph	NH_2	0.0125±0.00098	7.99	7.81–8.17
a16^{T}	N	NH	p-1-yl- 4-NO_2-Ph	2,4,6-TriMe-Ph	NH_2	0.0787±0.00141	8.1	7.77–8.44
a17^{T}	N	NH	p-1-yl- 4-CN-Ph	2,4,6-TriMe-Ph	NH_2	0.00566±0.00313	8.25	7.21–9.28
a18^{ET}	N	NH	p-1-yl - Pyridin-4-yl	2,4,6-TriMe-Ph	NHMe	0.00725±0.00119	8.14	7.83–8.45
a19^{T}	N	NH	p-1-yl - 4-COOEt-Ph	2,4,6-TriMe-Ph	NHMe	0.01666±0.00794	7.78	6.89–8.67
a20^{T}	N	NH	p-1-yl - 4-$CONH_2$-Ph	2,4,6-TriMe-Ph	NHMe	0.00461±0.0019	8.34	7.57–9.11
a21^{IT}	N	NH	p-1-yl - 4-SO_2NH_2-Ph	2,4,6-TriMe-Ph	NHMe	0.0069±0.00109	8.16	7.86–8.45
a22^{IT}	N	NH	p-1-yl -4-SO_2Me-Ph	2,4,6-TriMe-Ph	NHMe	0.00606±0.00204	8.22	7.59–8.85
a23^{T}	N	NH	p-1-yl - Pyridin-4-yl	2,4,6-TriMe-Ph	OMe	0.01332±0.00288	7.88	7.47–8.28
a24^{ET}	N	NH	p-1-yl - 4-COOEt-Ph	2,4,6-TriMe-Ph	OMe	0.01471±0.00704	7.83	6.94–8.73
a25^{ET}	N	NH	p-1-yl - 4-$CONH_2$-Ph	2,4,6-TriMe-Ph	OMe	0.00864±0.00149	8.06	7.74–8.39
a26^{T}	N	NH	p-1-yl - 4-SO_2NH_2-Ph	2,4,6-TriMe-Ph	OMe	0.01062±0.00258	7.97	7.52–8.43
a27^{T}	N	NH	p-1-yl - 4-SO_2Me-Ph	2,4,6-TriMe-Ph	OMe	0.0119±0.00454	7.92	7.21–8.64
a28^{T}	C-NO_2	O	4-CN-Ph	2,6-DiMe-Ph	H	0.58	6.24	N.R
a29^{T}	C-NO_2	O	4-CN-Ph	2,4,6-TriMe-Ph	H	0.056	7.25	N.R
a30^{T}	C-NO_2	O	4-CN-Ph	2,6-DiMe-4-CN-Ph	H	0.11	6.96	N.R
a31^{ET}	C-NO_2	O	4-CN-Ph	2,6-DiMe-4-Br-Ph	H	0.27	6.57	N.R
a32^{T}	-C-NO_2	O	4-CN-Ph	2,6-DiMe-4-Cl-Ph	H	0.16	6.8	N.R
a33^{T}	C-NO_2	O	4-CN-Ph	2,6-Di-OMe-Ph	H	4.28	5.37	N.R
a34^{IT}	C-NO_2	O	4-CN-Ph	2,6-DiCl-Ph	H	0.62	6.21	N.R
a35^{ET}	C-NO_2	O	4-CN-Ph	2,4,6-TriCl-Ph	H	0.17	6.77	N.R
a36^{T}	C-NO_2	O	4-CN-Ph	2,6-DiCl-4-NO_2-Ph	H	16.8	4.77	N.R
a37^{T}	C-NO_2	O	4-CN-Ph	2,4,6-TriBr-Ph	H	0.11	6.96	N.R
a38^{T}	C-NO_2	O	4-CN-Ph	2,6-DiBr-4-Me-Ph	H	0.034	7.47	N.R
a39^{T}	C-NO_2	O	4-CN-Ph	2,4,6-TriF-Ph	H	0.72	6.14	N.R
a40^{T}	C-NO_2	NH	4-CN-Ph	2,4,6-TriMe-Ph	H	30.17	4.52	N.R
a41^{T}	C-NO_2	NH	4-CN-Ph	2,6-DiBr-4-Me-Ph	H	3.85	5.41	N.R
a42^{T}	C-NO_2	NH	4-CH_3-Ph	2,4-DiBr-6-F-Ph	H	3.05	5.52	N.R
a43^{T}	C-NO_2	O	4-NO_2-Ph	2,6-DiMe-4-CN-Ph	H	0.72	6.14	N.R
a44^{ET}	C	O	4-CN-Ph	2,4,6-TriMe-Ph	-OMe	0.84±0.43	6.08	5.12–7.03
a45^{IT}	C	O	4-CN-Ph	2,6-DiMe-4-CN	-OMe	0.7±0.5	6.15	4.82–7.49
b46^{T}	C-Cl	O	p-Cl-Ph	2,4,6-TriMe-Ph	-	0.139±0.05	6.86	6.19–7.53
b47^{T}	C-Cl	O	p-Me-Ph	2,4,6-TriMe-Ph	-	0.201±0.04	6.7	6.30–7.10
b48^{T}	C-Cl	O	p-CN-Ph	2,4,6-TriMe-Ph	-	0.079±0.02	7.1	6.75–7.46
b49^{T}	C-Cl	O	p-CN-Ph	2,4,6-TriCl-Ph	-	0.155±0.04	6.81	6.30–7.32
b50^{T}	C-Cl	O	p-CN-Ph	2,4,6-TriBr-Ph	-	0.18±0.06	6.74	6.12–7.37
b51^{T}	C-Cl	O	p-CN-Ph	2,6-DiBr-4-Me-Ph	-	0.034±0.01	7.47	6.81–8.13
b52^{T}	C-Cl	O	p-CN-Ph	2,6-DiMe-4-Br-Ph	-	0.4±0.15	6.4	5.70–7.10
b53^{T}	C-Cl	O	p-CN-Ph	2,6-DiCl-Ph	-	0.22±0.05	6.66	6.22–7.09
b54^{T}	C-Cl	O	p-CN-Ph	4-CN-2,6-DiMe-Ph	-	0.144±0.03	6.84	6.47–7.22
b55^{IT}	C-Cl	O	p-CN-Ph	2,6-DiMe-Ph	-	0.54±0.21	6.27	5.54–6.99
c56^{T}	C=O	O	p-Cl-Ph	2,4,6-TriMe-Ph	H	0.78±0.17	6.11	5.70–6.51
c57^{T}	C=O	O	p-Me-Ph	2,4,6-TriMe-Ph	H	1.55±0.32	5.81	5.42–6.20
c58^{ET}	C=O	O	p-NO_2-Ph	2,4,6-TriMe-Ph	H	5.08±1	5.29	4.93–5.66
c59^{IT}	C=O	O	p-OMe-Ph	2,4,6-TriMe-Ph	H	1.59±0.21	5.8	5.55–6.05
c60^{T}	C=O	O	p-CN-Ph	2,4,6-TriMe-Ph	H	0.26±0.04	6.59	6.30–6.87
c61^{IT}	C=O	O	p-CN-Ph	2,4,6-TriBr-Ph	H	0.63±0.2	6.2	5.61–6.79
c62^{T}	C=O	O	p-CN-Ph	2,6-DiBr-4-Me-Ph	H	0.21±0.03	6.68	6.41–6.94
c63^{ET}	C=O	O	p-CN-Ph	2,6-DiMe-4-Br-Ph	H	1.19±0.24	5.92	5.55–6.30
c64^{T}	C=O	O	m- Cl -Ph	2,4,6-TriMe-Ph	H	14.4±4.21	4.84	4.30–5.39
c65^{T}	C=O	O	p-CN-Ph	2,6-DiMe-Ph	H	1.9±0.3	5.72	5.42–6.03
c66 ^{T}	C=O	O	p-CN-Ph	4-CN-2,6-DiMe-Ph	H	2.83±0.54	5.55	5.19–5.90
c67 ^{T}	C=O	O	p-CN-Ph	2,4,6-TriMe-Ph	Me	1.16±0.25	5.94	5.53–6.34
c68*^{T}	C=O	O	p-CN-Ph	2,4,6-TriMe-Ph	Me	3.36±0.99	5.47	4.92–6.02
d69^{ET}	C	O	4NO2-ph	2,4,6-TriMe-Ph	= O	1.4±0.3	5.85	5.45–6.25
d70^{T}	C	O	4-Me-Ph	2,4,6-TriMe-Ph	= O	1.4±0.4	5.85	5.32–6.39
d71^{T}	C	O	4-CN-Ph	2,4,6-TriMe-Ph	= O	0.37±0.1	6.43	5.93–6.94
d72^{T}	C	O	4-CN-Ph	2,6-DiMe-4-CN-Ph	= O	0.15±0.06	6.82	6.08–7.57
d73^{IT}	C	O	4-CN-Ph	2,6-DiMe-Ph	= O	1.4±0.17	5.85	5.63–6.08

a,b,c,d show the main skeletons of the compounds studied, (*) Ligand 68 has NMe instead of NH in the main skeleton; **R** is the carbon atom in the whole dataset except the structures 10 to 27, which is nitrogen atom; **Z** is carbon atom in the whole dataset except the structures 46 to 68 that is nitrogen; **p-1-yl** is piperidin-1-yl methyl; (—)

denotes no substitution in R_3 position; **ET** shows the external test compounds; **IT** represents the internal test compounds; **T** represents the compounds in the training set; **N.R** shows not reported and **interval** shows the 95% confidence intervals reported for the 3 replicate measurements of the pEC_{50} values.

Table 2
Molecular docking descriptors (MDDs) and their meanings.

No.	Symbol	Derived from	Meaning
1	BE	Docking	Estimated Free Energy of Binding
2	VE	Docking	$\sum$E(vdw + Hbond + desolv)
3	IE	Docking	Final Intermolecular Energy
4	EE	Docking	Electrostatic Energy
5	TI	Docking	Final Total Internal Energy
6	TE	Docking	Torsional Free Energy
7	UE	Docking	Unbound System's Energy
8	K_i	Docking	Inhibitory constant
9	pK_i	Docking	-log Ki

Table 3
The default values for the parameters of Lamarckian genetic algorithm (LGA).

Parameters	value
Population size	150
Maximum number of evals	2500000
Maximum number of generations	27000
Maximum number of top individuals	1
Rate of Gene mutation	0.02
Rate of cross over	0.8
GA cross over mode	Two-point
Mean of Cauchy distribution for gene mutation	0.0
The variance of Cauchy distribution for gene mutation	1.0
Number of generations picking worst individuals	10

3.5. Selection of the most significant descriptors

The variable selection and dimension reduction process were applied separately to the MDDs and the mixture of both functional groups and MDDs as a new ensemble of descriptors. For this purpose, the descriptors with constant and relatively constant values were deleted, then one of two descriptors with the pairwise correlation coefficient above 0.90 was also removed. The validation and test sets, each containing 11 compounds, were removed from the total data. By using (pEC_{50}) as a dependent variable and training data set descriptors as independent variables, the choice of the best structural or docking derived descriptors was carried out using the LASSO method in R software [40]. The descriptors correspond to the value of shrinkage parameter (λ_{min}) in which the cross-validation error is minimum was considered as the most significant variables. For comparative purposes, in the same way, SPSS 25 software [41] was used to select the most important variables using the stepwise regression method.

3.6. ANN modeling

A feed-forward ANN with backpropagation of error algorithm consisting of three layers (input, hidden and output) was used for the modeling of anti-HIV activity of studied compounds. A total of six structural based and docking-based molecular descriptors was given as input to the ANN and the experimental value of anti-HIV activity (pEC_{50}) was considered as the network output. A neural network with 4 neurons in the input layer, 4 neurons in the hidden layer and one neuron in the output layer were designed and its algorithm is written using the ANN toolbox in MATLAB 2015a software [42]. The ANN model was trained using the training series (51 compounds) with Bayesian- Regularization (BR) training function, the hyperbolic tangent sigmoid transfer function (tansig), and 2 training epochs. Then test series descriptors were given as inputs to the trained ANN and the equivalent activities were predicted using optimized values of network weights and biases.

4. Results and discussion

4.1. Descriptor generation

The MDDs were derived from the LR structure resulted in the inclusion of each studied compound, as the ligand, in the active site of the related receptors. To find such LR structures, molecular docking should be performed in the best docking condition to provide ligand-receptor structures with the best position of the ligand in the active site of the receptor. Thus, the extraction of docking derived descriptors was performed in three steps included determining the best molecular docking conditions by validation (dock-re-dock) procedure, docking the studied ligands in their related proteins structures in optimal conditions and finally extraction of descriptors by analyzing the output file of the molecular docking.

The validation technique is a self-docking procedure that cognate ligand is introduced into and out of the protein active site at different conditions and the best docking conditions can be achieved with regard to the value of binding energy, the number of clusters, the number of configurations in the best cluster and the run time. The validation process was performed in Autodock4.2 software as the following steps. First, the cognate ligand and formerly its matching protein (previously prepared according to Section 2.3) were opened in Autodock4.2 software. All hydrogen atoms were added to the crystallographic structure of the protein [43] and the non-polar hydrogen atoms attached to the carbon atoms were merged and Kollman charge was also added. Then the grid box parameters such as the distance between the grid points (spacing), the grid dimensions and coordinates adjusted in auto-grid. For this purpose, the grid with dimensions of 60 × 60 × 60 A° [44,45] was considered along the coordinate axes to include the protein's active site. The grid spacing was set at one-fourth of the length of the carbon-carbon single bond (carbon with sp^3 hybridization), which is approximately equal to 0.375 A°. Since the cognate ligands are in the active site of crystallographic structure, the coordinates of the grid center were adjusted according to the coordinates of the gravity center of the cognate ligand. In this regard, coordinates of the gravity centers of the cognate ligands were found (using ViewerLite5.0 software) to be 50.431, 63.194, 13.716 and 11.528, 14.168, 16.109 A^0 for 3M8Q and 3MEC receptors, respectively. Therefore, such coordinates were considered as grid center coordinates for self-docking studies. The molecular docking process is implemented using different algorithms such as the Lamarckian genetic algorithm (LGA), local search, and energy evaluation. Among them, the LGA is a more efficient method with more repeatable docking results compared to other algorithms [43] was used as a searching algorithm to find the best position of the ligand in the protein's active site. Different parameters affect the performance of the Lamarckian genetic search algorithm. One of them is the maximum number of energy evaluations, which usually has three levels of short, medium and long. The reported results [46–48] acknowledge that in most cases the average evaluation level is sufficient to be 2.500.000 (medium level) thus in all docking studies the maximum number of energy evaluations was set to medium level. The running number of the algorithm is another important factor to be estimated and optimized. Increasing the level of this factor will increase the accuracy of calculations, while also increasing the running time of molecular docking simulations. Therefore, the effect of this parameter on the accuracy of self-docking and also on the running time was investigated. For this purpose, parameters such as ligand displacement in space along the axis x, y and z (transition), the rotation in the three dimensional space around the x, y, and z-axis (quaternion), torsion and other conditions of the genetic algorithm and docking was adjusted to their default values in Autodock4.2 software (Table 3). The self-docking procedure was performed with the genetic algorithm running number of 100, 150 and 200 and parameters such as root mean

square deviation (RMSD) of ligand and protein binding, mean binding energies, number of clusters with multi conformations, number of conformations in the first cluster and run-time were calculated (Table 4). It has been established that successful docking is that the RMSD resulting from ligand and protein binding is less than 2 A°, the mean binding energy is more negative, there are few clusters while there is an accumulation of conformation in a cluster, and finally, the run-time is shorter [49]. Therefore, according to the facts and given the results in Table 4, a molecular docking simulation with the LGA implementation with run number of the 150 has a valid RMSD (resolution value < 2.0 A°), the lowest number of clusters, the highest number of conformations in the superior cluster and acceptable run-time were selected for further docking studies.

To obtain the best pose of the studied compounds (as ligands) and the best structure of the LR, the file of the compounds and the corresponding proteins, that were already prepared according to preparation method (Section 2.3), were opened in the Autodock 4.2 software. All settings related to the addition of Kollman charge and hydrogen atoms, definition of docking grid parameters were performed. Finally, under the best conditions obtained for molecular docking study (Table 4), each compound (as a ligand) was docked into the active site of the related protein (ligands 1–43 docked to 3M8Q and ligands 44–73 docked to 3MEC). The outputs of docking were saved as a dlg file for each molecule. The output files were analyzed and the LR conformation with the best pose of the ligand in the active site of the receptor was identified and 9 numerical descriptors, named docking derived descriptors, were extracted. These descriptors (are listed in Table 2) are mostly energy types. In addition to the docking derived descriptors, simple and interpretable structural descriptors can be derived from the structure of the studied compounds. Due to the fact that the functional groups and their numbers have a great influence on the interaction of drugs with the target, functional group descriptors (class 17 in Dragon software) were extracted for all compounds using Dragon 5.5 software. For this purpose, the optimized structure of compounds (Hyperchem output file (*hin)) was given as input to the Dragon and a total of 24 functional group descriptors were calculated. A mixture of these structural descriptors and the energy type docking derived descriptors were used in the development of the predictive QSAR model. Only 2 descriptors of functional group descriptors (Table 2) appeared in the final QSAR model.

4.2. Selection of relevant descriptors

Given that the main goal of this study was the simultaneous use of information on the interaction of studied azine compounds with proteins and some public structural descriptors in QSAR modeling, the interaction information of the ligand-receptor was extracted as energy descriptors from molecular docking and then they were mixed with functional group descriptors. In the ensemble of molecular docking derived and functional group descriptors (a total of 33 descriptors), non-significant or high correlated descriptors are probably presented that cause difficulty in the accurate modeling. Therefore, a preliminary data dimension reduction procedure (deleting descriptors and remaining one of two correlated descriptors) was performed. To find descriptors with the most important in the QSAR model, the validation and test data were removed from the data set and the training series data was only used in variable selection procedure using the LASSO method. In this process, descriptors were used as independent variables and pEC_{50} was selected as the dependent variable and the LASSO algorithm was run in R software and finally, only 10 descriptors, corresponded to λ_{min}, were selected as the most significant variables (Table 5).

Table 5
Descriptors selected among the mixture of MDDs and functional group descriptors using the LASSO method.

No	Symbol	Class	Meaning	β_i
1	**n135Triazine**	**Dragon Derived**	**Number of 1-3-5 Triazines**	**0.422**
2	**UE**	**Docking Derived**	**Unbound system Energy**	**0.221**
3	**pK_i**	**Docking Derived**	**(-Log) of Estimated Inhibition Constant, K_i**	**0.173**
4	**nCconj**	**Dragon Derived**	**Number of non-conjugated aromatic C (sp^2)**	**−0.129**
5	nNCN	Dragon Derived	Number of amidines derivatives	−0.105
6	nN	Dragon Derived	Number of Nitro group	−0.103
7	$NArNH_2$	Dragon Derived	Number of primary amines (aromatic)	0.059
8	nPyridazine	Dragon Derived	Number of Pyridazines	0.015
9	nCs	Dragon Derived	Number of total secondary C (sp^3)	0.004
10	VE	Docking Derived	vdW + Hbond + desolv Energy	−0.002

β_i means standardized coefficient.
The appeared descriptors in the final LASSO-BR-ANN model

Considering the small size of the data set and the fact that the LASSO method is used as the variable selection method for the first time, it was necessary to study the stability of the LASSO method in the variable selection. For this purpose, ten replicate variable selection were performed with LASSO so that at each time, the data set was randomly split into the training (51 compounds), validation (11 compounds), and test (11 compounds) sets and the training set was used in the variable selection. Variables corresponding to λmin (λ with the minimum 5-fold cross-validation error) were chosen as the most significant variables. The standardized regression coefficients were calculated and used as a criterion for the estimation of the importance order of variables. Table 6 shows the first five important selected variables in each run along with their importance order (importance decreases from left to right). The results obtained show that the number and type of the variables selected in all runs are completely identical and in only one case the order of importance for two variables (nCconj and nNCN) is moved. Therefore, it can be verified that the LASSO method has desired stability in the variable selection in terms of type and importance of the selected variables."

In order to evaluate the efficiency of only MDDs and only functional group descriptors in creating predictive QSAR models, the same descriptor selection process was also performed to the MDDs and/or functional group descriptors. In each case, only 4 descriptors were

Table 4
The dock-re-dock results for different proteins obtained from the different numbers of LGA running.

Parameters	Dock-re-dock results					
	P1			P2		
No. of LGA running for protein	100	150	200	100	150	200
RMSD[a] (Å)	0.44	0.69	0.51	0.62	0.61	0.63
No. of Clusters	13	7	17	10	5	12
No. of conformations in 1st cluster	28	75	54	43	134	104
Lowest binding energy (KJ mol^{-1})	−11.69	−12.20	−11.76	−11.64	−11.93	−11.83
Running time (min)	74	116	142	34	55	140

[a] RMSD: root mean square deviation, P1 and P2 corresponded to 3M8Q and 3MEC receptors respectively.

Table 6
Selected descriptors at different runs by the random splitting of the data set.

No. of runs	Selected descriptors (arranged by importance order)
1	n135Triazines, TI, pKi,nCconj, nNCN
2	n135Triazines, TI, pKi, nNCN, nCconj
3	n135Triazines, TI, pKi, nCconj, nN
4	n135Triazines, TI, pKi, nCconj, nN
5	n135Triazines, TI, pKi, nCconj, nN
6	n135Triazines, TI, pKi, nCconj, nN
7	n135Triazines, TI, pKi, nCconj, nN
8	n135Triazines, TI, pKi, nCconj, nN
9	n135Triazines, TI, pKi, nCconj, nN
10	n135Triazines, TI, pKi, nCconj, nN

selected as the most important descriptors (Table 7) and used in QSAR modeling.

4.3. ANN modeling and optimization of ANN parameters

One of the most important steps in QSAR studies is to find a model that establishes a proper relationship between the descriptors and the activities of the compounds studied. Among different modeling methods, ANN as a nonlinear method has been considered by researchers [50]. The most important steps in ANN modeling is the design of ANN structure, determination of training data sets, determining the initial values of the weights, network training, and simulation of the network response [14]. The training process requires a series of examples, which includes the ANN inputs and their corresponded target values. During training, weights and biases are adjusted to minimize the mean square error (MSE) as an efficiency function of ANN. ANN models often encounter problems such as over-fitting, over-training and inappropriate prediction power of the model. The published QSAR studies suggest using the Bayesian Regularization (BR) training function (TRAINBR in Matlab toolbox) to solve these problems [25,26,51–54]. The benefits of ANN models trained with BR training function are less over-training and do not require the validation set, however, the use of the validation set increases the reliability of the model. On the other hand, it has been mathematically proven that the BR training function creates a consistent model so that the test set is not required even when the number of data points is low [55]. Therefore, in this study, a feed-forward ANN with a back-propagation BR training algorithm was selected. Each network has at least one input layer, an output layer, and several hidden layers. The number of hidden layers is often selected by try and error process. However, ANN models with one hidden layer have been proposed for computational chemistry studies [34,56]. As a result, ANN with one hidden layer was used and its algorithm was written using the MATLAB software toolbox. For developing an ANN model with low prediction error, ANN parameters such as the number of input variables (number of descriptors), hidden layer nodes, training epochs and the type of transfer function (Hyperbolic tangent sigmoid (tansig), Logarithmic sigmoid (logsig) were optimized. Dataset was randomly divided into training, validation and test sets. The series of training data was used to train and optimize the values of weights and biases, The validation set was used to prevent over-fitting via early stopping process [55] and the test set was used for validation of the final and optimized ANN model. Although according to the benefits of the BR training algorithm, virtually no validation set required but validation set was used to further assurance in the lack of over-fitting during the optimization process. During the training process, the mean square error (MSE) was calculated for the validation set. The minimization of MSE value was considered as an optimization goal. MSE was calculated according to the following equation:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2 \tag{4}$$

where $\hat{y}_i$ and y_i are the predicted and actual activity values, respectively.

There is no order of importance among the descriptors selected by LASSO. On the other hand, in the optimization of ANN inputs, the descriptors are given in the order of importance as inputs, therefore, the order of descriptor importance should be specified. Thus, the standardized coefficients of descriptors were calculated according to the LASSO regression model and their magnitudes were considered as a criterion of significance and the data series was re-arranged according to the importance of descriptors (Table 5). The following equation was used to calculate the standardized coefficients.

$$\beta_i = \hat{\beta}_i \frac{s_x}{s_y} \tag{5}$$

Where β_i and $\hat{\beta}_i$ are the standardized and unstandardized regression coefficients, respectively. s_x and s_y are the standard deviation of descriptor x and response y, respectively.

To find the optimal conditions of the ANN model, parameters such as the number of inputs (descriptors), number of nodes in the hidden layer, number of training epochs and the type of transfer function were simultaneously optimized. For this purpose, BR-ANN models with one hidden layer and the different number of inputs (2–10) and nodes in the hidden layer (2–10) and the transfer function of logarithmic sigmoid and hyperbolic tangent sigmoid were designed. All ANN models were trained using the training set then used for the prediction of biological activities of the validation set. The minimization of MSE value for the prediction of biological activities of molecules in the validation set was considered as the optimization goal. The optimal conditions for the superior LASSO-BR-ANN models with different transfer functions were summarized in Table 8. Therefore, the LASSO-BR-ANN model with 4 mixed descriptors (n135triazine, UE, pKi, nCconj) and hyperbolic tangent sigmoid transfer function, under the optimal conditions of other parameters (Table 8) was recognized as the superior model of this study.

For the comparison purposes, ANN models were constructed using the BR training algorithm and hyperbolic tangent sigmoid transfer function in the hidden layer. The designed ANN models were optimized the same as before, except that only four descriptors of the MDDs alone or only the structural descriptors were used as inputs. The results showed that the ANN models with architectures of 4–10-1 (4 training epochs) and 4-9-1 (45 training epochs) are the optimal models based on the structural descriptors and MDDs, respectively. All constructed models were

Table 7
Descriptors selected among the MDDs or functional group descriptors using the LASSO method.

Descriptor category	Symbol	Meaning	β_i*
MDDs only	UE	Unbound system Energy	0.477
	EE	Electrostatic Energy	0.240
	pK_i	-Log(Estimated Inhibition Constant, Ki)	0.143
	VE	(vdW + Hbond + desolv) Energy	−0.038
Functional group counts only	n135Triazines	Number of 1-3-5 Triazines	0.460
	nNCN	Number of amidines derivatives	−0.194
	nCs	Number of total secondary C (sp3)	0.187
	Nn	Number of Nitro group	−0.119

Table 8
The MSE values for the best ANN models constructed using the mixture of MDDS and functional group descriptors.

No. Descriptors	Training function	Transfer function	No. Node	epochs	MSE
6	BR	logsig	8	8	0.044
4	BR	tansig	4	2	0.041

evaluated to determine the superiority of modeling using the combination of descriptors compared to the modeling using descriptors alone. For this purpose, three ANN models (constructed by the use of structural descriptors alone, MDDs alone and their mixture) were applied for the prediction of pEC_{50} values for the compounds in the validation set under the resultant optimal conditions and architecture. The MSE values were obtained as 0.07, 0.18 and 0.04 for the models constructed by the use of the functional group descriptors alone, only MDDs and their mixture, respectively. The results obtained clearly proved the superiority of the mixture of the functional group descriptors and MDDs in the QSAR modeling and also confirmed the adequacy of the information behind the mixture of descriptors, as a new class of descriptors, in the explanation of the biological activities. Therefore, the ANN model using the mixture of the functional group descriptors and MDDs was adapted as the superior model for further studies.

4.4. Validation of model

4.4.1. Prediction errors

The validity and accuracy of a QSAR model are usually confirmed with acceptable errors in the prediction of the biological activities of new compounds (external test set). Therefore, the proposed LASSO-BR-ANN model based on the mixture of MDDs and functional group descriptors was applied for the prediction of anti-HIV inhibitory activities (pEC_{50} values) of the compounds in the external test and validation (internal test) sets under the optimal conditions. The predicted values to gather with the actual values and their corresponded 95% confidence intervals (based on the experimental data) are given in Table 9. Regarding the results obtained, it is clear that the prediction errors are low (lower than 10%) and about 90% of the predicted pEC_{50} values are in the experimental confidence interval (CI), which confirms the lack of systematic error and thus the accuracy of the proposed LASSO-BR-ANN model. Further studies were performed by plotting the residual values against the real values. The results obtained (Fig. 2) show that the residual values are symmetrically distributed around the horizontal axis, which again indicates the absence of systematic error in the predicted results.

4.4.2. Statistical parameters

The accuracy of the proposed model (LASSO-BR-ANN) was also evaluated by the calculation of statistical parameters for the model predicted pEC_{50} values. The calculated statistical parameters with their mathematical equations are given in Table 10. In order to evaluate the performance of the proposed model, the calculated statistical parameters were compared with their acceptable values or adequate criteria. Some of such criteria are as follows: The coefficient of determination (R^2) for the test/validation set should be greater than 0.6 [57]. The coefficient of determination for linear regressions with intercepts forced to zero, which are obtained by plotting the predicted values versus actual values (R_0^2) or

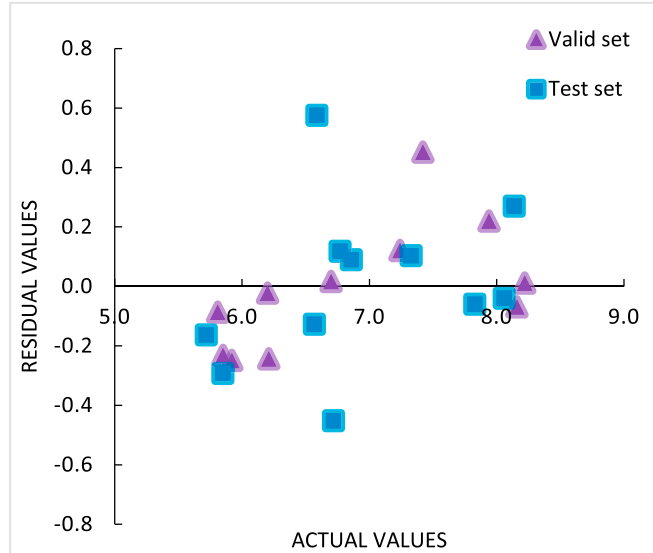

Fig. 2. Residual plot for the internal and external test sets.

plotting the actual values versus predicted values ($R_0'^2$) should be as close as to R^2, while the ratio of their difference to R^2 should be less than 0.1 and their absolute difference should be less than 0.3 [58]. At least one of the slopes of the regression lines with forced zero intercepts (k or k' Table 10) should be in an acceptable range of 0.85–1.15. Also, the values of R^2_m and R'^2_m should be greater than 0.5 [59]. The agreement of the calculated statistical parameters with the accepted values (Table 10) confirmed the validity, reliability, and accuracy of the proposed LASSO-BR-ANN model.

4.4.3. Y- randomization test

The Y-randomization technique was used to investigate any random relationship between descriptors and biological activities in the dataset. For this purpose, ten response vectors in which the main biological activities are replaced with random values were produced and used as the dependent variable in QSAR modeling. BR-ANN models were trained in the optimal conditions using randomized biological activities and then used to predict the activities of the compounds [60]. The values of R^2 obtained from ten runs of the Y-randomization test are shown in Table 11. The small values of Y-randomization correlation coefficients indicate the absence of random relation between independent and dependent variables of the training data in the developed LASSO-BR-ANN model.

Table 9
The results obtained for the prediction of pEC_{50} values for compounds in internal (validation) and external test sets usning the proposed LASSO-BR-ANN model.

Test set				Valid set			
Actual	predicted	%Error	CI[a]	Actual	predicted	%Error	CI
6.72	7.17	6.73	6.33–7.11	7.42	6.97*	−6.09	7.28–7.56
7.33	7.23	−1.40	6.89–7.76	7.24	7.12	−1.67	6.34–8.14
8.14	7.87	−3.31	7.83–8.45	7.94	7.72	−2.77	7.85–8.02
7.83	7.89	0.77	6.94–8.73	8.16	8.23	0.84	7.86–8.45
8.06	8.10	0.50	7.74–8.39	8.22	8.21	−0.13	7.59–8.85
6.57	6.70	1.94	N.R[b]	6.21	6.45	3.92	N.R
6.77	6.65	−1.74	N.R	6.70	6.68	−0.24	6.30–7.10
6.86	6.77	−1.30	5.12–7.03	5.81	5.90	1.50	5.42–6.20
6.59	6.01*	−8.73	4.93–5.66	6.20	6.22	0.36	5.61–6.79
5.72	5.88	2.86	5.55–6.30	5.92	6.17	4.21	5.55–6.30
5.85	6.14	5.01	5.45–6.25	5.85	6.08	3.97	5.63–6.08

[a] 95% confidence interval for experimental values.
[b] Not reported.

Table 10
Statistical parameters for the proposed LASSO-BR-ANN model obtained by the prediction of pEC50 values for compounds in internal and external test sets.

NO.	Formula	Internal test et	External test set	Acceptable range/value
1	$\mathrm{PRESS} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$	0.46	0.78	-
2	$\mathrm{SEP} = \sqrt{\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n}}$	0.21	0.27	-
3	$\mathrm{MAE} = \frac{\sum_{i=1}^{n}\lvert y_i - \hat{y}_i\rvert}{n}$	0.16	0.21	-
5	$\mathrm{REP}(\%) = \frac{100}{\bar{y}} \times \sqrt{\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n}}$	2.96	3.82	-
5	$\mathrm{MSE} = \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n}$	0.04	0.07	-
6	$\mathrm{MRE} = \frac{\sum_{i=1}^{n}\left\lvert\frac{y_i - \hat{y}_i}{y_i}\right\rvert}{n} \times 100$	2.33	3.11	-
7	$R^2 = \frac{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$	0.96	0.88	>0.6
8	R_0^2(predicted vs actual(intercept is zero))	0.94	0.87	Close to R^2
9	Relative $R_0^2 = \frac{(R^2 - R_0^2)}{R^2}$	0.014	0.013	>0.5
10	$R_m^2 = R^2 \times \left[1 - (R_0^2 - R^2))^{\frac{1}{2}}\right]$	0.83	0.79	Close to R^2
11	$R_0^{'2}$(actual vs predicted (intercept is zero)	0.95	0.88	>0.5
12	Relative $R_0^{'2} = \frac{(R^2 - R_0^{'2})}{R^2}$	0.009	0.000	<0.3
13	$R_m^{'2} = R^2 \times \left[1 - (R_0^{'2} - R^2))^{\frac{1}{2}}\right]$	0.87	0.87	
14	R-R = $\lvert R_0^2 - R_0^{'2}\rvert$	0.023	0.014	
15	k = Slope predicted vs actual(intercept is zero)	0.999	0.999	$0.85 \le k \le 1.15$
16	k' = Slope actual vs predicted (intercept is zero)	1.000	1.000	$0.85 \le k' \le 1.15$

1-Predictive Residual Sum of Squares, 2-Standard Error of Prediction, 3-Mean Absolute Error, 4- Relative Error of Prediction, 5- Mean Square Error, 6- Mean Relative Error. y_i is observed (experimental) value, $\hat{y}_i$ is predicted value and $\bar{y}$ is the average value of observed values, *p* is the number of descriptors and n is the number of compounds. R^2 squared correlation coefficient between the observed and predicted value of compounds with intercept R_0^2 the squared correlation coefficient between the observed and predicted value of compounds without intercept. $R_0^{'2}$ Bears the same meaning as R_0^2, but uses the reversed axis.

4.4.4. Diversity analysis

Diversity analysis indicates the distribution of compounds in all series including training, validation and test sets. The distribution in the chemical space should be such that the test and validation series represent the whole data. In this regard, Euclidean distances were calculated for two compounds of P_i and Q_j using their descriptor values according to the following equation.

$$r_{ij} = \left\|P_i - Q_j\right\| = \sqrt{\sum_{k=1}^{n}\left(p_{ij} - q_{ij}\right)^2} \tag{6}$$

where p_{ij} and q_{ij} are descriptors values of the compound. The mean Euclidean distances of each compound ($\bar{r}_{ij}$) was calculated and normalized between zero and one. In order to display the distribution of all data, the normalized values were plotted versus biological activities (pEC_{50}). The results (Fig. 3) shows the proper distribution of training set data across a wide range of chemical space that proves the stability of the model. Due to the proper distribution of test and validation data in the interval of train set data, the predictability of the model is also verified [61].

4.4.5. Applicability domain

The applicability domain approach is used to examine the predictability power of the developed QSAR model and determines the outlier observations among the train and test sets. For obtaining the applicability domain, the hat diagonal matrix (H) which is a projection matrix should be calculated [62]. The hat matrix is known as the projection matrix and its elements is called leverage (h_i). The hat matrix (73×73) was calculated by the projection of observations (y) on the predicted values ($\widehat{y}$) according to the following equation:

$$h_i = x_i(X^TX)^{-1}x_i^T \quad i = 1,\ 2,\ \ldots,\ n \tag{7}$$

where x_i is the row vector of descriptors for each studied compound, X is the training set descriptors matrix and T represents the transpose of the defined vector or matrix. The elements of the hat matrix are called leverage (h_i), which shows the effect of a particular observation on predicted values with respect to its position in the chemical space of all inputs [62]. The leverage values were used to obtaining the applicability domain and detection of any outlier compounds in the data set. For this purpose, the values of the standardized residuals of the training and test series were plotted versus the leverage values extracted from the corresponding hat matrix. The results obtained (Fig. 4) show William's plot which is a display of the applicability domain. As shown in Fig. 4, all compounds lie in the applicability domain and any outlier was not observed. In obtaining the applicability domain, the critical value for leverage (h*) was considered equal to 3p/n, where p is the number of descriptors in the model plus one and n is equal to the number of compounds in the train set [63]. With taking p and n equal to 4 and 62

Table 11
R^2 values resulting from 10 times random permutations of the responses in the Y-randomization test.

Run	1	2	3	4	5	6	7	8	9	10
R^2	0.009	0.133	0.006	0.025	0.070	0.078	0.174	0.073	0.099	0.005

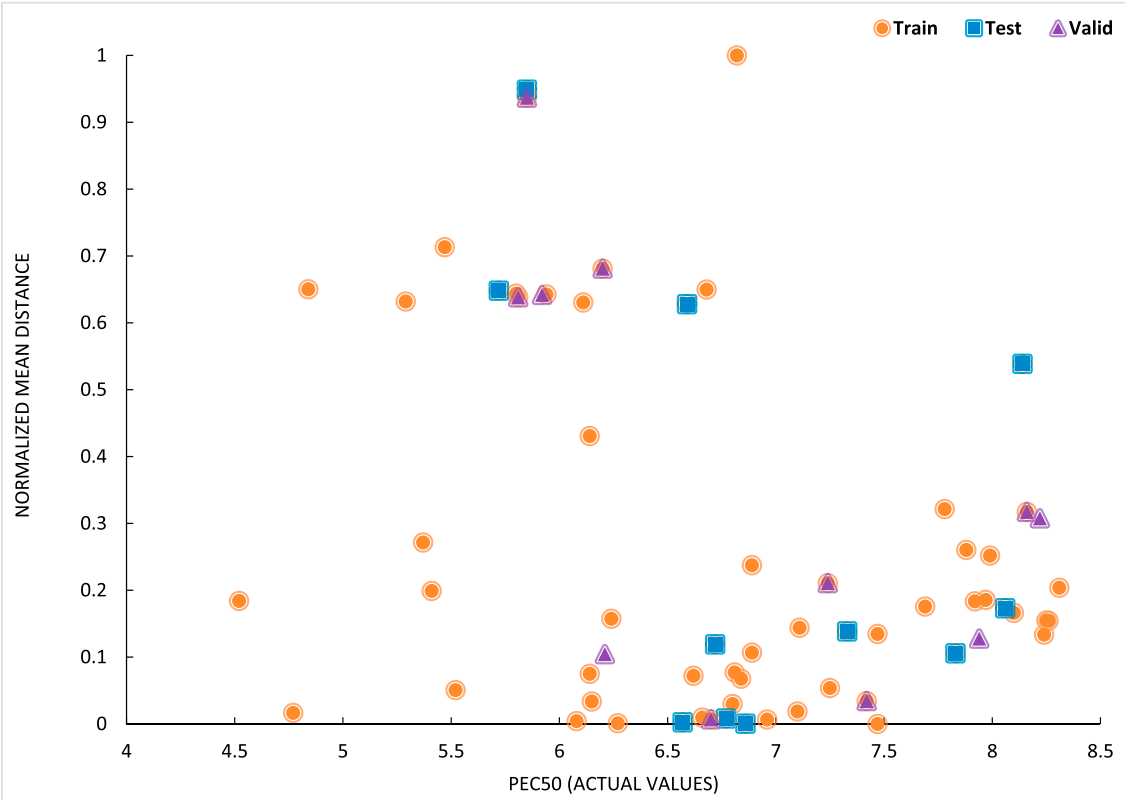

Fig. 3. The result obtained from the diversity analysis.

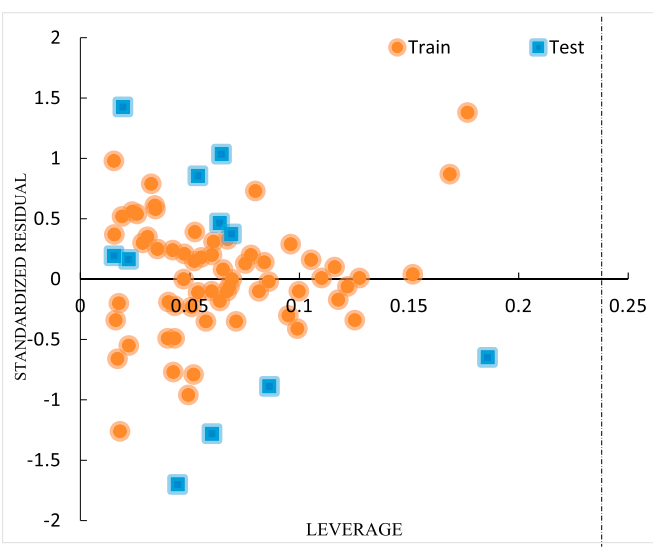

Fig. 4. The plot of the applicability domain for the proposed LASSO-BR-ANN model (h* = 0.24).

respectively, h* was found to be 0.24, which also specified in William's plot (Fig. 4). It is clear that no compound has a value of h greater than 0.24, consequently, the predicted biological activity values for the training and test sets are reliable and the developed LASSO-BR-ANN model as a QSAR model is credible enough [64].

To prove the validity of the calculated AD, especially when dealing with the new external test set, the ADs of the model were repeatedly computed ten times using the ten different training and test sets derived from the random splitting of the data set (Section 4.2). A number of 10 AD graphs were obtained by the use of the selected variables (Table 5) and the corresponding training and test sets. The results obtained (Fig. S4 to Fig. S13) show that the data in all test sets are within the acceptable range of the calculated ADs and no outliers are observed.

4.4.6. Brief description of selected descriptors

The variables selected by LASSO were investigated to detect any probable existence of co-linearity. For this purpose, the value of the variance inflation factor (VIF) was calculated for each independent variable using the below equation [65]:

$$VIF = \frac{1}{1 - R_i^2} \; . \; i = 1.2.3 \cdots . p \tag{8}$$

where p is the number of descriptors and R_i^2 is the square of multiple correlation coefficient that is obtained from a regression of x_i on all of the other predictor variables. It has been established that the values of VIF in the range of 0.1–10 confirm the lack of co-linearity [65]. According to results obtained (Table 12), there is no considerable co-linearity between the descriptors selected by the LASSO method. The correlation between the descriptors used in the development of the BR-ANN model was also investigated.

The contribution percentages of descriptors appeared in the final LASSO-BR-ANN model were estimated using the following equation [66]:

$$C_i = 100 \frac{\Delta m_i}{\sum_{i=1}^{N} \Delta m_i} \tag{9}$$

where C_i is the contribution percentage of descriptor i, N is the number of model descriptors (equal to 4) and Δm_i is the mean absolute error obtained from predicted activities values using a model constructed with the participation of descriptor i when its values are randomly perturbed in the range of own main values.

According to the results obtained (Fig. 5), the order of the

Table 12
VIF values for the descriptors appeared in the final LASSO-BR-ANN model.

Descriptors	n135Triazines	UE	pKi	nCconj
VIF	3.32	3.44	1.16	1.09

contribution percentages of descriptors in the final model is n135triazine> UE> pKi> nCconj.

The most effective descriptor is the presence of the triazine ring (n135triazine) in the structure of the studied compounds. Its positive effect (Table 5) shows that the presence of the triazine ring increases biological activity. This is due to the presence of NH_2 groups in the structure of most compounds with a triazine ring. The presence of such substitution increases the structural similarity of these compounds to the reference structure of Etravirine, as well as the interactions with the amino acids in the structure of target protein. In order to investigate the interactions in the presence and absence of the triazine ring, the complex structures derived from the interaction of compounds 2 and 13 with the target protein was analyzed using Ligplot 3 software (Fig. S1). (It should be noted that the compounds have been investigated by Ligplot software v3.0. The two structures are overlapped with each other, the circle around the amino acids shows that they are those amino acids that have hydrophobic or hydrophilic interaction with both structures. An amino acid which its color is red (no circle around it), is related to the amino acid corresponded to the bold structure, in this example means structure named 13. Green dashed lines represent for hydrophilic bonds). The interaction information of the ligand-receptor, summarized in Table 13, clearly shows that the number of surrounding amino acid, the hydrophobic and the hydrogen bonds for the compound 13 are much greater than the compound 2. Therefore, the biological activity of 13 ($pEC_{50} = 8.24$) is greater than that of 2 ($pEC_{50} = 7.42$).

The second most effective descriptor of the model is UE with a positive effect and introduces the effect of the binding energy into the model. As UE value is more positive, the binding free energy of the ligand-receptor becomes more negative. This means the stronger interactions have occurred and the activity increased. For example, we can compare the interaction information of compounds 30 and 35 (Table 13). As can be seen, compound 35 (with less activity than 30) with more negative unbonded energy has fewer connections with the adjacent amino acids (Fig. S2), less negative binding free energy and thus has less activity compared to compound 30. It should be noted that the interpretation of the effect of a descriptor with considering the ligand-receptor interactions is very difficult, especially in cases where the values of other descriptors are not constant. For example, a more positive UE value may not increase the biological activity due to the lack of triazine ring and/or small pK_i value. Therefore, we tried to select comparative constituents so that when the effect of a descriptor is surveyed, the values of other descriptors are almost constant.

The other significant descriptor is pK_i, which is related to the free energy according to the following equation [39]:

$$K_i = \exp(\Delta G \times 1000) / (R_{cal} \times T) \quad (10)$$

Where ΔG is the docking binding free energy, R_{cal} is gas constant (1.987 Cal mol^{-1} K^{-1}) and T is 298 K. This descriptor has a positive effect factor in the model and increases the biological activity of the compounds. The positive effect of pK_i is due to this that compounds with a larger theoretical inhibitory coefficient (pK_i) have more possibility to establish hydrophobic and hydrophilic bond interactions. Comparison of the interactions of the complexes of compounds 4 and 33 with the receptor (Fig. S3 and Table 13) shows that compound 4 with larger pK_i than that of Compound 33 have the larger number of surrounding amino acids, hydrophilic and hydrophobic interactions.

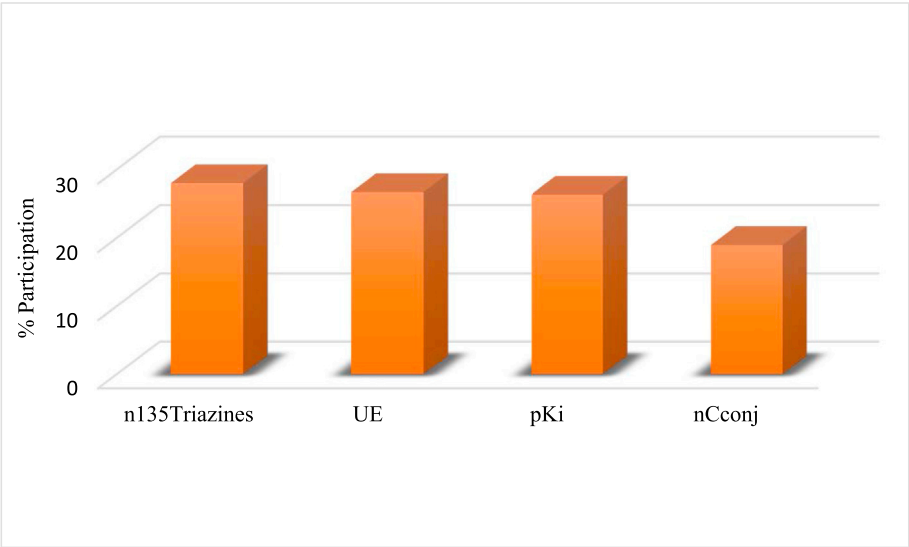

Fig. 5. Participation percentage for descriptor appeared in the proposed LASSO-BR-ANN model.

Table 13
The Interactions of several studied molecules with the amino acids in the active site of the corresponding protein and the number of surrounding residues.

No.	Descriptors values				Interactions		Residues	pEC_{50}
	n135Triazines	UE	pK_i	nCconj	Hydrophobic	Hydrophilic		
2	0	−0.66	8.38	0	52	0	12	7.42
13	1	−0.77	8.89	0	54	3	18	8.24
30	0	−1.07	8.16	0	42	1	14	6.96
35	0	−1.2	8.17	0	39	1	13	6.77
4	0	−1.38	9.29	0	62	2	17	6.89
33	0	−1.38	7.11	0	42	1	14	5.37

5. Conclusion

The QSAR model was developed to predict the anti-HIV activity of a series of azine compounds. The descriptors were calculated using molecular docking and Dragon software. After reducing the dimensions of the introduced descriptors using the LASSO method, selected descriptors were defined as inputs of ANN. The model has proper statistical parameters such as low MSE and the high coefficient of determination(R^2) equal to 0.07 and 0.88 for the test set and 0.04 and 0.96 for the validation set. According to the results of this research, choosing a suitable method for the selection of the most relevant descriptors is so crucial for the development of the QSAR model. The QSAR model developed using descriptors selected by stepwise regression has greater prediction error than the superior predictive model introduced in this study. It should be noted that the MDDs alone do not have enough ability to make a good relationship with biological activity, and adding different descriptors could improve the relationship between structure and activity, hence using the functional group descriptors that have simple interpretation and significant influence in establishing different interactions are recommended. This study is the first report for constructing a neural network model with descriptors selected using LASSO method.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Zeinab Mozafari: Methodology, Software, Writing - original draft, Investigation, Writing - review & editing. **Mansour Arab Chamjangali:** Supervision, Writing - review & editing, Data curation. **Mohammad Arashi:** Software, Validation, Writing - review & editing.

Acknowledgment

The authors are thankful to the Shahrood University of Technology Research Council for supporting this work.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chemolab.2020.103998.

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